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Probabilistic Combinatorics SS24

Sheet 8

Due: 10. June at 19:00

(1) (exercise 2, sheet 3 revisited) Let again $G = ([n], E)$ an undirected graph and $F_1, ..., F_n \subseteq F$ a collection of possible colors for every node, i.e. F is a superset of colors and F_i is the set of colors allowed for the vertex i. Here let $\#F_i = 8r$ for every $i \in [n]$, $r > 0$. Further for every $c \in F_i$, let there be at most r neighbors j of i, s.t. c is also allowed for j, i.e.

$$\#\{j \in N(i) \mid c \in F_j\} \leq r.$$

Show: There is an allowed coloring f (i.e. no edge is monochromatic) of G, with $f(i) \in F_i$.

(2) A hypergraph $H = (V, E)$ is a possible generalization of a graph: Instead of $E \subseteq \binom{V}{2}$ consider a subset $E \subseteq 2^V$. We call the elements of E hyperedges.

An allowed 2-coloring of a hypergraph $H = (V, E)$ is a map $f : V \to [2]$, s.t. $\#f(e) = 2$ for all $e \in E$.

Let $H = (V, E)$ be a hypergraph, such that every hyperedge $e \in E$ connects at least k nodes (i.e. $\#e \geq k$ for every $e \in E$) and every hyperedge e intersects at most d other hyperedges.

Show: If $e(d + 1) \leq 2^{k-1}$, then there is an allowed 2-coloring of H.

(3) Let $G = (V, E)$ with $E \subseteq V \times V$ a directed graph. Let

$$\delta(G) = \min_{v \in V} \#\{w \in V \mid (v, w) \in E\}$$

the minimal out-degree of G and

$$\Delta(G) = \max_{v \in V} \#\{w \in V \mid (w, v) \in E\}$$

the maximal in-degree of G.

Show: If there is a $k > 1$, s.t. $e \cdot (\Delta(G) \cdot \delta(G) + 1)(1 - 1/k)^{\delta(G)} < 1$, then there is a $i \geq 1$, such that G contains a simple cycle of length ik.

A simple cycle of length ℓ in a graph is a sequence $e_1, ..., e_\ell$ of edges, such that $e_i = (v_i, v_{i+1})$, $1 \leq i \leq \ell - 1$ and $e_\ell = (v_\ell, v_1)$ and the vertices $v_1, ..., v_\ell$ are pairwise distinct.

(4) An (undirected) graph $G = (V, E)$ is called d-regular, if every node is contained in exactly d edges. If $G = (V, E)$ is a d-regular graph and $V = V_1 \cup ... \cup V_r$ is a partition of V (i.e. $V_i \cap V_j = \emptyset$ for $i \neq j$) with $\#V_i \geq 2ed$, then there is a $V' \subseteq V$ with $\#(V' \cap V_i) = 1$ for every i and there are no edges between nodes of V' (i.e. $\binom{V'}{2} \cap E = \emptyset$).